

0.1 Solution

To determine the maximum possible value of g , we construct distinct positive integers a_1, a_2, a_3 such that g divides $\frac{P}{a_i} + 1$ for each i , where $P = a_1 a_2 a_3$, and their sum is strictly less than 3×10^6 .

0.1.1 Key Observations and Construction

1. **Necessary Condition:** For g to divide $\frac{P}{a_i} + 1$ for all i , all a_i must be congruent modulo g . Let $a_i \equiv c \pmod{g}$. Then $P \equiv c^3 \pmod{g}$, and the condition $\frac{P}{a_i} \equiv -1 \pmod{g}$ implies $c^2 \equiv -1 \pmod{g}$.

2. **Maximizing g :**

- Choose $g = c^2 + 1$ for the largest integer c such that the sum of $a_1 = c$, $a_2 = c + g$, and $a_3 = c + 2g$ is less than 3×10^6 .
- The sum is $c + (c + g) + (c + 2g) = 3c + 3g = 3(c + g)$. Substituting $g = c^2 + 1$, the constraint becomes:

$$3(c + c^2 + 1) < 3 \times 10^6 \implies c^2 + c + 1 < 10^6.$$

- The largest integer c satisfying this is $c = 999$, since $999^2 + 999 + 1 = 999001 < 10^6$, while $c = 1000$ yields $1000^2 + 1000 + 1 = 1001001 > 10^6$. Thus, $g = 999^2 + 1 = 998002$.

3. **Validation:**

- The integers are $a_1 = 999$, $a_2 = 999 + 998002 = 999001$, and $a_3 = 999 + 2 \times 998002 = 1997003$. Their sum is $999 + 999001 + 1997003 = 2997003 < 3 \times 10^6$.
- **GCD Condition:** For any a_i , $\frac{P}{a_i} + 1$ is divisible by g . For example, with $a_1 = 999$:

$$\frac{P}{a_1} + 1 = (999001 \times 1997003) + 1.$$

Since $g = c^2 + 1$, we have $c^2 \equiv -1 \pmod{g}$. Thus:

$$(c + g)(c + 2g) + 1 \equiv c \cdot c + 1 \equiv c^2 + 1 \equiv 0 \pmod{g}.$$

Similar calculations hold for a_2 and a_3 . The greatest common divisor of all three terms is exactly g , as shown by the pairwise coprimality of the cofactors $c + 1$, $2c + 1$, and $2g + 3c + 1$. For $c = 999$, these cofactors are 1000, 1999, and 1999002, with GCD 1.

4. **Maximality:**

- For any $g' > 998002$, the minimal sum with $k = 3$ would exceed 3×10^6 . For example, $c = 1000$ yields $g = 1000001$, with sum $3(1000 + 1000001) = 3003003 > 3 \times 10^6$.
- Other constructions with $k > 3$ or different residues yield smaller g due to stricter sum constraints.

Thus, the maximum possible g is 998002 .